
Introduction to Web Development



Course Overview

Introduction

- ▶ Web Design and Programming (AC2070)
- ▶ Structure:
 - ▶ Lectures
 - ▶ Assignments
 - ▶ Project
 - ▶ Final exam
- ▶ Tools:
 - ▶ NodeJS
 - ▶ A good ~~text~~code editor:
 - ▶ [VS Code](#) / Sublime Text / Atom / Brackets
 - ▶ A standard-compliance web browser:
 - ▶ Chrome / Firefox / Edge / Safari

Course Objectives

- ▶ At the end of this class, you will be able to:
 - ▶ Design and implement webpages using HTML
 - ▶ Make stylistic decisions with CSS
 - ▶ Create interactive webpages with JavaScript
 - ▶ Create dynamic websites with ExpressJS
 - ▶ Enhance interactive websites with AJAX
- ▶ And beyond:
 - ▶ Understand the client-server programming model and apply this to your designs
 - ▶ Speak the web programming lingo
 - ▶ Have fun with web development, and enjoy creating websites!

Contents

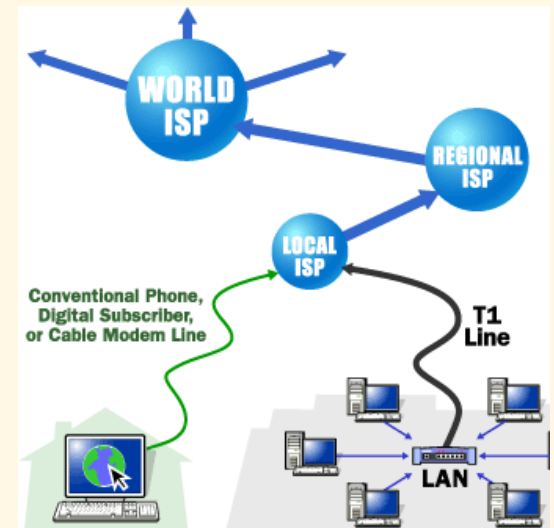
- ▶ HTML language
 - ▶ Learn to compose web pages to present contents
- ▶ CSS (Cascading Style Sheets)
 - ▶ Learn to polish web pages for better presentation
- ▶ Client-side programming
 - ▶ Learn to make web pages interactive with client-side JavaScript
- ▶ Server-side development
 - ▶ Learn to program dynamic web sites with server-side JavaScript

WWW (World Wide Web)

Definition



- ▶ WWW = The Web
 - ▶ A system of globally accessible documents and resources
 - ▶ A collection of websites and webpages stored in web servers, that are interlinked by URLs
- ▶ Not to be confused with Internet
 - ▶ Internet is a network
 - ▶ WWW is a service on Internet



History of WWW

- ▶ 1989: WWW (including HTTP, HTML) invented by Timothy Berners-Lee
- ▶ 1991: Mosaic, first popular web browser
- ▶ 1994: WWW Consortium (W3C) at MIT
- ▶ 1995: JavaScript → Interactive and dynamic webpages
- ▶ 1996: Google, the most popular search engine
- ▶ 2000: Burst of “dot-com” bubble
- ▶ 2002: Rise of blogs, wikis → Web 2.0 = era of user-generated content (blogs, wikis)
- ▶ 2003: Rise of social networks (MySpace, Facebook,...)
- ▶ 2005: Rise of web applications with AJAX
- ▶ 2014: HTML5 (location, 2D/3D graphics, media,...)



Demystification

- ▶ Webpage (or just “page”): A document which can be displayed in a web browser
- ▶ Website (or just “site”): A collection of grouped webpages and made available at a certain address
- ▶ Web server: A computer (server) that hosts websites
 - ▶ Apache, IIS, Tomcat, nginx, ExpressJS,...
- ▶ Web browser: A software that retrieves and displays webpages to users
 - ▶ Opera, Chrome, Edge, IE, Netscape, Safari,...

Basic Components of WWW

- ▶ Content: Information displayed by the web browser
 - ▶ Static: Content that doesn't change
 - ▶ Dynamic: Content that varies based on various factors
- ▶ Instructions: What the browser needs to do
 - ▶ Formatting: How the content is to be displayed
 - ▶ Navigation: How to travel between webpages
 - ▶ Others: Pop-up windows, validation, customization
- ▶ **Build a website = creating Content and Instructions**

Web Standards

- ▶ Web browsers and servers need to speak the same “languages”
- ▶ URL (Uniform Resource Locator): An address that helps to locate a webpage
- ▶ HTML (HyperText Markup Language): The markup (formatting) language for webpages
- ▶ HTTP (HyperText Transfer Protocol): A protocol (set of communication rules) allowing web browsers to retrieve resources from web servers
- ▶ CSS, JavaScript, JSON, XML,...

Web Development

What is It?

- ▶ **Building** and **maintenance** of websites
 - ▶ Make websites work
 - ▶ Make websites work fast
 - ▶ Make pages look great
- ▶ What do web devs need?
 - ▶ HTML
 - ▶ CSS
 - ▶ JavaScript

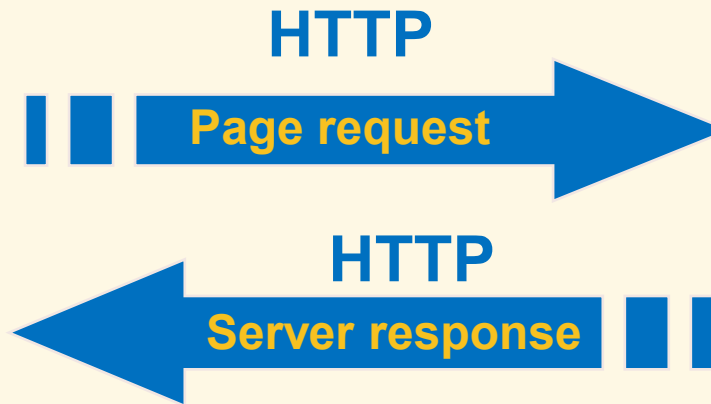


Front-end and Back-end Development

**Client running a
Web Browser**



**Server running
Web Server Software**



Front-end development:

- HTML
- CSS
- Client-side scripting

Back-end development:

- Web server
- Database



**Full-stack
development**

Web Applications

- ▶ Internet is more and more ubiquitous, with very high-speed and wireless connections
- ▶ Web browsers are nowadays equipped with powerful features (Web 2.0)
 - ▶ HTML5, CSS3, AJAX, REST, graphics (canvas, WebGL, SVG), hardware accessibility (camera, location, notification),...
- ➔ Web is now one of important application/service development and distribution platforms (besides server, desktop, mobile)
 - ▶ No installation needed → many advantages to users and providers
 - ▶ Ex: Google Maps, Gmail, MS Office 365, web games,...



SOA

▶ Software distribution with Package Model has many limitations

- ▶ Low compatibility with user devices → high software development cost
- ▶ Feature, patch updates
- ▶ Difficult to guarantee user activities
- ▶ Difficult to assess usability, user experience

→ Service Oriented Architecture (SOA)

- ▶ Service in place of application
- ▶ A service may target end users, or other services
- ▶ 4P: Process, Practice, People, Platform
- ▶ Ex: map, payment, storage (cloud), authentication, AI, advertisement, user data analytics,...

